

Product Evaluation - Live Translate

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- **Date:** 2026-07-11
- **Video demo:** <https://www.loom.com/share/4ff5f53052d849429fcdf5e2cc805a6b>
- **LLM provider / model:** Google Gemini · gemini-3.1-flash-lite
- **Backend target:** local gateway `http://localhost:8787` · **deployed gateway** `https://naveed-lt-gw.fly.dev`

Verdict

*This is shippable. The backend is a clean two-service split - a public Node gateway and a **private** Python AI service that the browser can never reach directly - and it does the one hard thing well: identical (text, target) never hits the LLM twice, and a cache hit is **~57x faster** than a miss (11 ms vs 626 ms p95). Translations are natural Mexican Spanish with correct regional register (tapetes , not the Castilian alfombras) and prices/percent/SKU codes preserved verbatim. Errors fail loud (LLM failure → 502 , logged), so it never ships English while looking healthy. The live test against homedepot.com's real homepage strings passed end-to-end through the deployed gateway. **Strongest part:** caching correctness + observability (one request id greps across both services). **Weakest part:** the cost figures below use the provided sla.json placeholder Anthropic prices (left unedited on purpose - benchmark/ is marked do-not-edit), so the dollar amounts are conservative; real Gemini Flash-Lite pricing is materially lower, making the true savings larger.*

Rubric score (from eval/report.json): 70 / 70 auto (+ 30 manual, grader-scored)

Auto breakdown: Widget lights up 15/15 · Caching 20/20 · Performance & SLA 15/15 · Logging & observability 10/10 · Service separation & contract 10/10.

1. Performance & cost (from benchmark/bench.py , cold cache)

Metric	Result	SLA	Pass?
Cache hit p95	11.1 ms	≤ 60 ms	✓
Cache miss p95	626 ms	≤ 3500 ms	✓
Cache hit rate	75.0 %	≥ 60 %	✓
Throughput	1268 req/s	≥ 20	✓
Error rate	0.0 %	≤ 1 %	✓
Cost per miss	\$0.000161	-	-
Monthly savings from cache	\$60.36	-	-

`python benchmark/bench.py` **exits 0** - all five SLAs met. Hit vs miss speedup ≈ **57x**.

Cost caveat (read this): `benchmark/sla.json` still carries the provided placeholder prices (anthropic / claude-sonnet-4-6 , \$3/\$15 per Mtok). I left that file untouched because AGENTS.md marks benchmark/ as do-not-edit. The actual provider here is **Gemini Flash-Lite**, whose published input/output rates are far lower, so the real cost-per-miss and monthly bill are **lower** than

shown and the savings from caching are **larger**. Treat the dollar column as a conservative placeholder, not the true Gemini bill.

(An earlier `eval.py` / `bench.py` pass ran against a fully-warm cache and reported 100% hit rate / \$0 cost - a degenerate run. The table above is from a **fresh cold cache**, which is the honest measurement.)

2. Live-website test

- **Site tested:** <https://www.homedepot.com> (live homepage - a real site I don't control)
- **Method:** pulled 8 real visible strings from the live DOM, then translated them by calling the **deployed** gateway (`https://naveed-lt-gw.fly.dev/translate/batch`) **from homedepot.com's own page origin** - the exact cross-origin path the widget uses.
- **Did it translate? Yes** - HTTP `200` , all 8 real homepage strings returned as natural es-MX through the deployed product. (This exercises the translation + CORS + cache path. A full visual DOM page-flip on a strict-CSP site is done via the **Chrome extension's background worker** by design - that's what the 60-90s video captures.)
- **CORS / CSP:** the gateway's `Access-Control-Allow-Origin: *` let homedepot's origin call it, and homedepot's CSP `connect-src` did **not** block the fetch. Note: injecting the widget `<script>` itself is blocked by homedepot's `script-src` (expected) - which is exactly why the extension exists and the console loader is only for permissive pages.
- **Batch speed:** page translation goes through `/translate/batch` , which fans its texts out to the LLM **concurrently** (bounded semaphore), not one-at-a-time. A fresh 20-string batch through the **deployed** gateway completes in **~1.3 s** total (vs ~12 s if run sequentially).
- **Cache on re-translate:** re-ran an identical batch → **all cached: true** , server latency effectively **0 ms** (client ~80-320 ms, mostly network RTT to `sjc`). Cache proven on real content.
- **Resilience:** `200` from the real origin, **0 errors**; on an LLM failure the service returns `502` (never the untranslated English).
- **Screenshots:** before/after visuals are in the submitted video (the extension flipping the live page to es-MX, then the cache-hit badge on re-translate).

Sample translations (real homedepot.com homepage strings)

Original (EN)	Translation (es-MX)	Numbers/codes kept?	OK?
UP TO 35% OFF Select Furniture, Rugs & Home Decor	HASTA 35% DE DESCUENTO en muebles, tapetes y decoración seleccionados	✓ 35%	✓
Shop All Savings	Ver todas las ofertas	-	✓
Today Only! Fast Free Delivery	¡Solo por hoy! Envío rápido y gratis	-	✓
Air Conditioners	Aires acondicionados	-	✓
Patio Furniture	Muebles de exterior	-	✓
Building Materials	Materiales de construcción	-	✓
Log In	Iniciar sesión	-	✓
Daily Deals	Ofertas del día	-	✓

Additional preservation check (workload): Add to cart – only \$1,299.00 for model SKU-4471, save 20%! → Agregar al carrito – solo \$1,299.00 por el modelo SKU-4471, ¡ahorra un 20%! - \$1,299.00 , SKU-4471 , 20% all preserved; Agregar (es-MX) not Añadir (es-ES).

Multi-language (stretch goal), same English → different targets through the deployed gateway: Sign in → es-MX Iniciar sesión · pt-BR Entrar · fr Connexion .

3. Dimension scorecard

Dimension	Pass / Partial / Fail	Evidence
Translation accuracy	✓ Pass	8/8 live homedepot strings + workload render accurate, fluent es-MX
Mexican-Spanish register (es-MX)	✓ Pass	tapetes (not alfombras), Agregar/gratis (not Añadir/gratuito); es-ES correctly differs
Numbers / prices / codes preserved	✓ Pass	\$1,299.00, SKU-4471, 20%, 35%, \$50 all verbatim
Page coverage	✓ Pass	Batch translated every string sent; full DOM walk handled by the widget/extension in the video
Cache effectiveness	✓ Pass	57× local speedup; 8/8 cache hits on real content; SQLite survives restart (verified)
Latency vs SLA	✓ Pass	All 5 SLAs green; bench.py exits 0
Error handling (no silent English)	✓ Pass	LLM failure → 502 + logged; no input-as-output fallback anywhere
Resilience on a real site	✓ Pass	200 from homedepot origin, 0 errors; CSP behavior understood and handled by the extension
UX polish	✓ Pass (provided)	Widget/extension are the provided, unmodified frontend; backend adds no UX regressions

Observability: /stats reports an accurate hit rate; gateway /health nests the AI service's health; both services log one structured JSON line per request; a single X-Request-Id greps end-to-end across gateway.log and ai-service.log (verified with both a client-supplied and an auto-generated id).

Deploy & hygiene: both services on Fly.io - gateway **public** (https://naveed-lt-gw.fly.dev , /health green), AI service **private** (no public IP, reached only via naveed-lt-ai.internal:8000), SQLite on a persistent volume. .env / *.db / *.log / node_modules / .venv git-ignored; no edits to widget/ · extension/ · benchmark/. Stretch goals shipped: **multi-language target** (any BCP-47 code) and **docker-compose.yml** (docker compose up runs both, verified end-to-end).

4. Top fixes before shipping

1. **Own the cost model.** Move the price table out of the provided benchmark/sla.json (do-not-edit) into a config you control, set to Gemini Flash-Lite's real published rates, so the cost/savings numbers reflect the actual provider instead of the Anthropic placeholder.

2. **Friendly limits.** Add per-IP rate limiting on the gateway returning `429` ; today an overload or error surfaces as the widget's generic "can't reach backend" message.
 3. **In-flight de-dup.** Batch translation now fans out concurrently (this fixed a sequential-batch bottleneck that made a full page take ~60 s). The next refinement is coalescing concurrent identical misses, so a burst of the same new string in one batch makes one LLM call, not N.
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What to submit

- **This file** (`PRODUCT_EVAL.md` , or a PDF export) **+ the 60–90s video** showing the extension translating a real page into Mexican Spanish and a cache hit in the badges.
- Add the video URL at the top once recorded (currently `TODO`).
- No **Fail/Partial** rows to fix - the only caveats are the cost placeholder (item 1) and the video URL.